



Inspiring Excellence

Multi-Task Deep Learning Framework for Unified Cattle Health and Behavior Monitoring

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1. ABSTRACT

This thesis proposes a unified, lightweight, spatiotemporal multi-task deep learning framework for cattle monitoring from image and video inputs. It simultaneously addresses Body Condition Scoring (BCS), behavior recognition, lameness detection, and individual cow identification using only publicly available datasets. The model employs a shared EfficientNet-B0 backbone with CBAM attention and task-specific heads, with temporal modeling via LSTM for sequence analysis. The framework theoretically reduces duplicated backbone memory by about 75% compared with deploying four separate backbone networks, supporting edge-oriented cattle monitoring.

2. RESEARCH OBJECTIVES

- 1 Build a unified multi-task framework for four cattle monitoring tasks.
- 2 Use publicly available datasets to improve transparency and reproducibility.
- 3 Develop a lightweight, edge-deployable architecture using shared feature learning.
- 4 Compare single-task and multi-task performance and analyze generalization.

3. PROBLEM STATEMENT

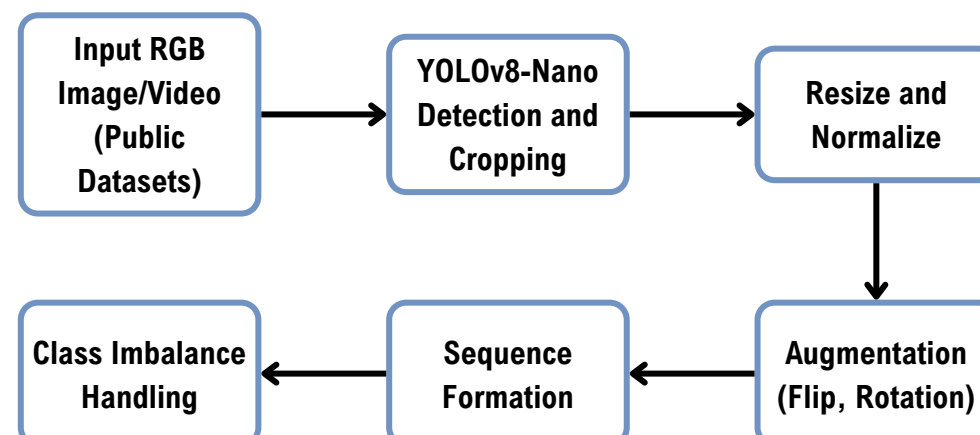
- Existing studies treat cattle monitoring tasks in isolation.
- Many rely on private datasets, limiting reproducibility.
- Cross-dataset evaluation is rarely performed.
- Models often require high-end GPUs or separate task-specific pipelines, making edge deployment difficult.

➔ A unified, low-cost, vision-based framework is needed to support practical cattle health and behavior monitoring using public datasets.

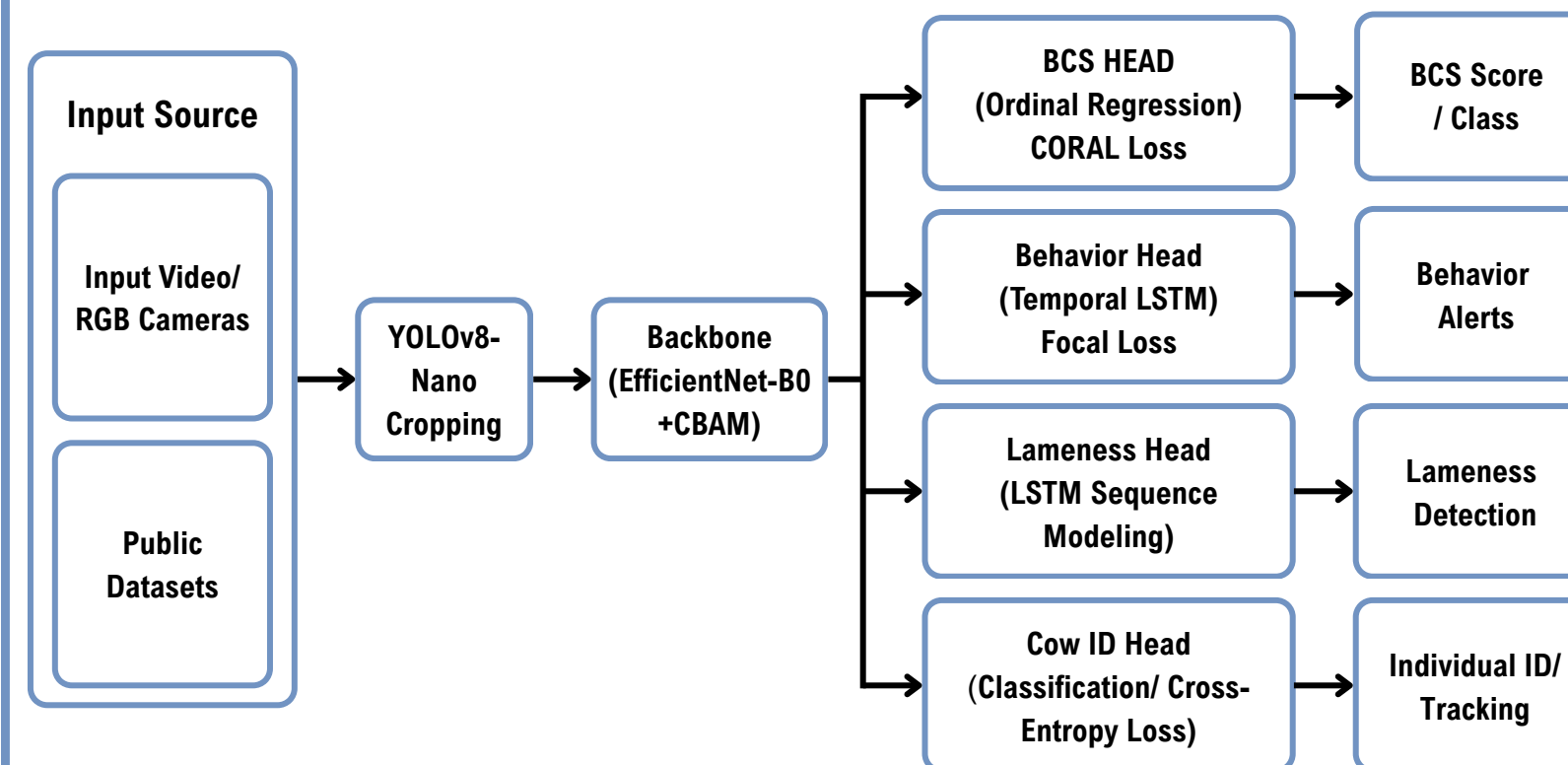
4. DATA COLLECTION & PREPROCESSING

Dataset	Content	Size/Details
ScienceDB	Body Condition Scoring (BCS)	53,566 images
Dryad	Depth Grayscale Edge (DGE) for BCS	5,923 images
MmCows	Behavior recognition (7 classes, severe imbalance)	213,686 cropped frames, capped to ~21,000 balanced training samples
CattleLameness	Locomotion / lameness analysis	50 video clips
OpenCows2020	Cow identification	4,736 images

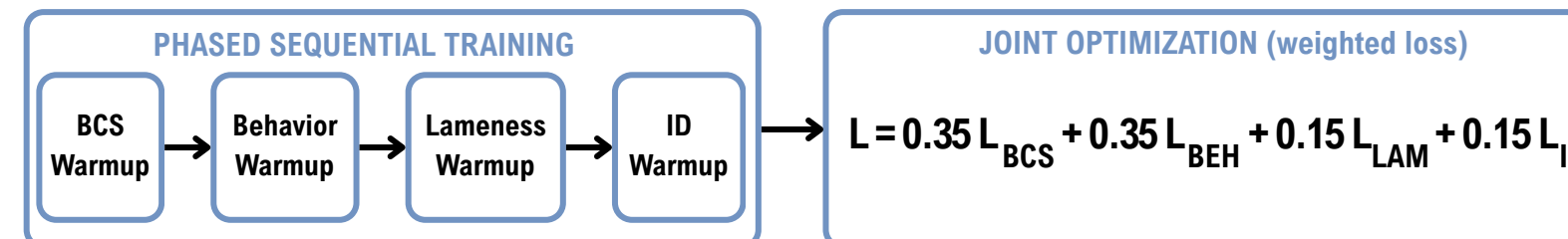
Preprocessing Pipeline:



5. PROPOSED METHODOLOGY / FRAMEWORK



Training Strategy



Framework Benefits



6. BACKBONE SELECTION / MODEL TRAINING

Backbone (Model)	BCS (MAE)		Behavior (Macro F1)	Lameness (AUC)	Cow ID (Top-1)	Params (M)
	Dryad(DGE)	ScienceDB				
ResNet-18	0.8675	0.5800	0.7134	0.7200	45.56%	11.2
MobileNetV3-Small	0.5250	0.7090	0.6810	0.7228	78.83%	2.5
ResNet-50	0.6300	0.6485	0.7037	0.7444	53.02%	23.5
DenseNet121	0.5875	0.6292	0.7366	0.7944	82.46%	7.0
EfficientNet-B0	0.6175	0.5566	0.7445	0.9829	86.49%	5.3

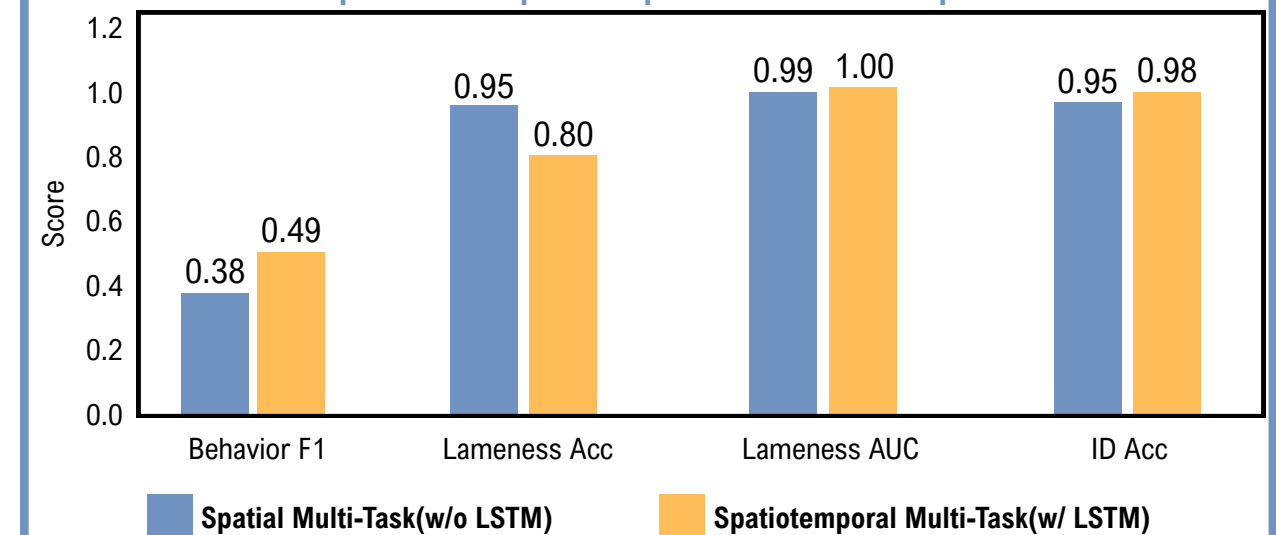
EfficientNet-B0 was selected as the shared backbone because it provided the best overall trade-off across task performance and model size, with only 5.3M parameters.

7. RESULTS & VISUALIZATION

Key Performance Summary Table:

Task	Single-Task Best Results (Best backbone)	Spatiotemporal Multi-Task Results (Proposed Framework)
BCS (MAE)	0.5566 (ScienceDB)	0.7827
Behavior (Macro F1)	0.7445 (EfficientNet-B0)	0.4948
Lameness	93.05% Accuracy 0.9829 AUC (Spatial Baseline)	80.00% Accuracy* 1.0000 AUC* (Threshold-tuned)
Cow ID (Top-1 Acc)	86.49% (EfficientNet-B0)	97.58% (Spatiotemporal)

Spatial VS. Spatiotemporal Multi-Task Comparison



*Post-hoc threshold-tuned result; not valid for final evaluation because it uses test-set feedback.

8. KEY ANALYSIS

- EfficientNet-B0 achieved the best overall single-task trade-off.
- Temporal modeling improved cow ID and gave preliminary lameness sequence results.
- CORAL loss improved BCS: MAE 0.5566 vs 0.6940.
- Focal Loss improved behavior F1: 0.7445 vs 0.7074.
- Multi-task learning reduced memory duplication but caused negative transfer.
- CBVD-5 testing revealed strong domain shift (F1 = 0.1245).

9. CONCLUSION

This work presents a public-dataset, reproducibility-oriented multi-task pipeline for unified cattle health and behavior monitoring. The model is designed for edge-oriented deployment and shows strong cow identification performance, while lameness sequence results remain preliminary. Future work should address negative transfer using gradient balancing methods such as GradNorm or PCGrad and improve multi-farm generalization.

10. REFERENCES

- [1] Tan, M., & Le, Q. V. (2019). *EfficientNet: Rethinking model scaling for convolutional neural networks*. Proceedings of the 36th International Conference on Machine Learning (ICML).
- [2] Woo, S., Park, J., Lee, J. Y., & Kweon, I. S. (2018). *CBAM: Convolutional Block Attention Module*. Proceedings of the European Conference on Computer Vision (ECCV).
- [3] Andrew, W., Greatwood, C., & Burghardt, T. (2021). *Visual identification of individual Holstein-Friesian cattle via deep metric learning*. Computers and Electronics in Agriculture, 185, 106133.